Field	Entry
Source ID	WBC-0052
Title	Design and Assessment of Ergonomics of Hand Tools Based on Gender-Specific Operating Strategy: Two Case Studies on Shovels and Pruning Shears
Source type	Thesis (academic)
Link / File	https://etda.libraries.psu.edu/catalog/12041
Craft applicability	Bamboo (root extraction using shovel, spade and axe)
Primary Why-theme	7. Posture, Grip and Body Mechanics
Process stage	Sub-step 1.2: Extraction of Bamboo Root
Questions answered	Why is the labor-intensive digging and extraction of bamboo roots physically demanding, and how does shovel design affect musculoskeletal load?
Evidence level	E3
Relevance note	Directly measures biomechanical and physiological load during shovel-based digging work, providing scientific grounding for the report's own description of root extraction as labor-intensive and time-consuming.